# Respiratory Disease Prediction from Multi-modal Data: A Report for Data, Model, Performance and Uncertainty Analysis

by AutoHealth

**Abstract**

This report details the experiments conducted by AutoHealth, a multi-agent system designed for antonymous health data modeling. This task is to predict respiratory disease using a combination of structured tabular metadata (demographics, symptoms, history) and respiratory audio recordings (cough and vowel sounds). A dual-branch late-fusion neural network was developed, integrating features from both modalities. To enhance robustness and provide uncertainty estimates, a deep ensemble of three models was trained with controlled architectural variations. The final model achieved a Macro-F1 score of 0.8384 on the test set, demonstrating strong generalization. Comprehensive uncertainty analysis revealed that incorrect predictions are associated with higher model uncertainty, providing a valuable tool for identifying low-confidence cases in clinical deployment. The report includes a detailed data analysis, model training methodology, performance evaluation, and actionable insights for clinical application.

## 1 Data Analysis

### 1.1 Dataset Overview

The dataset is a multi-modal collection designed for respiratory disease classification. It comprises 546 unique candidates, split into training (437), validation (54), and test (55) sets. Each candidate is represented by two data modalities:

- **Tabular Metadata**: An 11-column CSV file containing demographic information (age, gender), symptom/history indicators (e.g., wheezingHistory, phlegmCough, familyAsthmaHistory), smoking exposure (packYears), and the target label (disease: Class 0, 1, or 2).
- **Audio Recordings**: Two WAV files per candidate: a cough recording (avg. 16.0s) and a vowel recording (avg. 6.1s), both sampled at 48 kHz. Precomputed audio embeddings (JSON files) are also provided, representing deep features extracted from the audio signals.

The primary evaluation metric is Macro-F1, which is suitable for handling the observed class imbalance.

### 1.2 Feature Analysis and Clinical Correlations

Analysis revealed strong predictive signals within the data, aligning with clinical intuition.

- **Tabular Features**: Age and packYears (smoking exposure) showed high point-biserial correlation with the disease label (0.567 and 0.372, respectively). Categorical features like wheezingHistory and phlegmCough were also highly significant (Chi-square $p < 1e\text{-}07$). For instance, Class 1 had an 84.7% positive rate for wheezingHistory, compared to only 5.3% for Class 0.
- **Audio Features**: After mean-pooling the variable-length cough embeddings, 206 out of 512 feature dimensions showed significant inter-class differences (ANOVA $p < 0.05$). These audio features exhibited low correlation with strong tabular features ($|r| < 0.1$), suggesting they provide complementary, non-redundant information for classification.

### 1.3 Data Quality and Challenges

Several data quality issues were identified and addressed in the preprocessing pipeline:

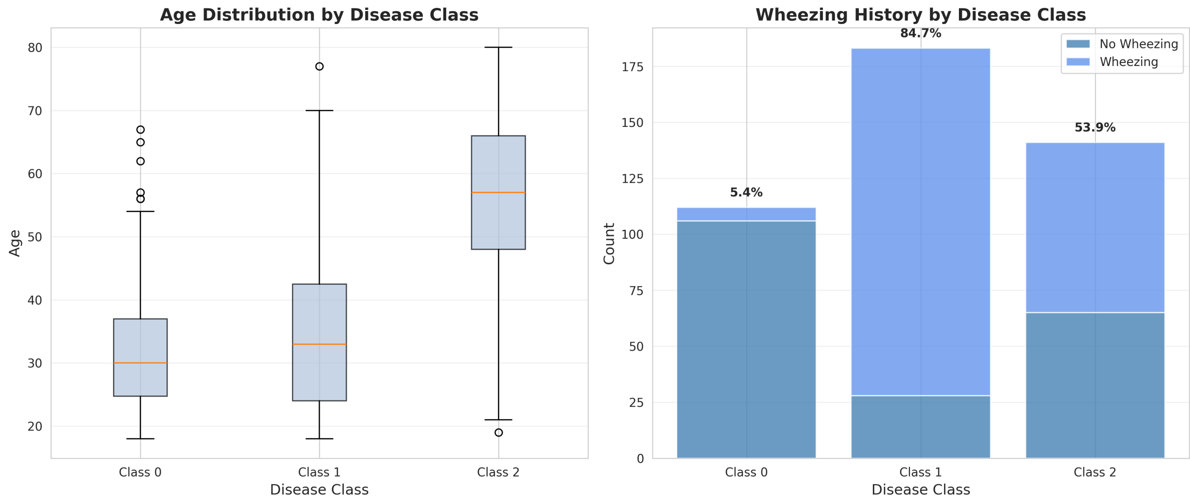


Figure 1: Relationship between key features and the target disease class. **Left:** Age distribution across disease classes, showing Class 0 as youngest and Class 2 as oldest. **Right:** Prevalence of wheezing history within each class, highlighting its strong association with Class 1. These features were prioritized during model design.

- **Missing Values**: The 'coldPresent' column contained significant missing values (115 in train set). These were imputed using the mode from the training data.
- **Class Imbalance**: The target distribution is uneven. In the training set, Class 1 is most frequent (183 samples), followed by Class 2 (141), and Class 0 (113). This imbalance was actively mitigated during training.
- **Audio Embedding Inconsistency**: The precomputed embeddings had variable sequence lengths (e.g., (1,512), (4,512)). A global average pooling operation was applied to standardize them into fixed-length vectors.
- **Distribution Shift**: Kolmogorov-Smirnov tests indicated a mild distribution shift in some audio embedding dimensions between the training and validation/test sets, necessitating careful feature scaling.

# 2 Model Training

## 2.1 Data Preprocessing

A comprehensive preprocessing pipeline was implemented to prepare the multi-modal data:

- **Tabular Data**: Categorical variables were label-encoded. Missing values in 'coldPresent' were filled with the training set mode. A simple interaction feature, 'elderly_with_wheezing', was created. Numerical features ('age', 'packYears') were standardized using the training set's mean and standard deviation.
- **Audio Data**: For each candidate, the 'emb_cough.json' and 'emb_vowel.json' files were loaded. The variable-length sequences were mean-pooled into 512-dimensional vectors each, then concatenated to form a 1024-dimensional audio feature vector, which was subsequently standardized.
- **Class Imbalance Mitigation**: Two strategies were employed: 1) Class weights ([1.25, 0.77, 1.00]) were calculated for the loss function to penalize errors on minority classes more heavily. 2) A 'WeightedRandomSampler' was used during training to ensure each mini-batch had a balanced representation of all three classes.
- **Data Augmentation**: To improve generalization, Gaussian noise was added to both tabular and audio feature vectors during training (std=0.03 and 0.02, respectively).

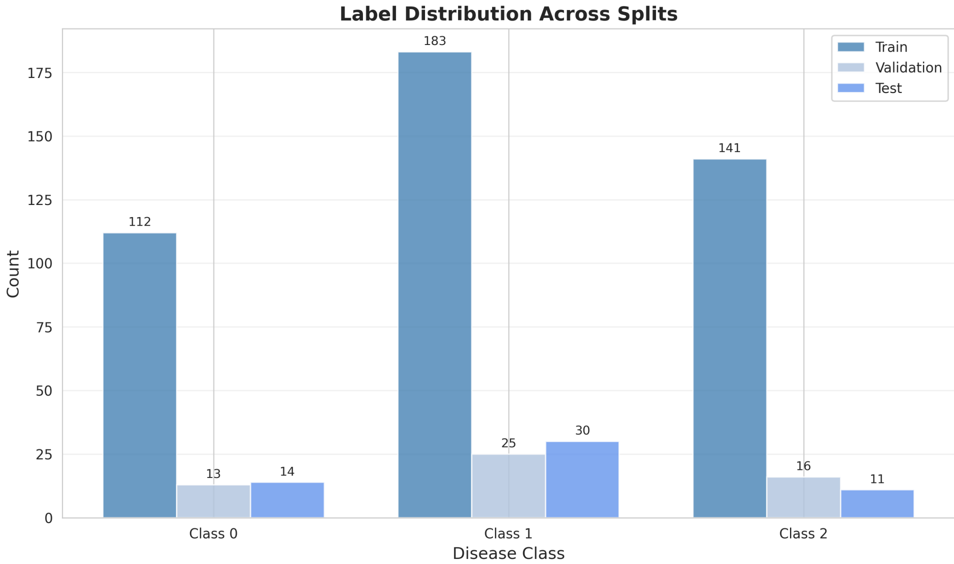


Figure 2: Label distribution across training, validation, and test splits. The class imbalance (Class 1 > Class 2 > Class 0) is consistent across splits, justifying the use of class-weighted loss and balanced sampling strategies during model training to prevent bias.

## 2.2 Model Architecture

A custom dual-branch late-fusion neural network was designed to effectively combine information from both modalities.

- **Audio Branch**: A Multi-Layer Perceptron (MLP) takes the 1024-dim audio vector, projects it through layers (1024→512→256) with BatchNorm and Dropout, outputting a 256-dim representation.
- **Tabular Branch**: A separate MLP processes the 10-dim tabular vector through layers (10→64→32), outputting a 32-dim representation.
- **Fusion & Classification**: The outputs from both branches are concatenated (288-dim). A final classification head (MLP: 288→128→64→3) produces the logits for the three disease classes.

This architecture allows each branch to learn specialized representations before combining them for the final decision.

## 2.3 Hyperparameters

Key hyperparameters were selected to balance learning capacity and regularization:

- **Optimizer**: AdamW with initial learning rate $lr = 5 \times 10^{-4}$ and weight decay $5 \times 10^{-4}$.
- **Learning Rate Schedule**: ReduceLROnPlateau scheduler (factor=0.5, patience=10 epochs).
- **Regularization**: Dropout rates of 0.4 in key layers, combined with BatchNorm, were used to prevent overfitting.
- **Training**: Batch size of 32 for up to 80 epochs, with early stopping (patience=20) based on validation loss.
- **Ensemble**: Three independent models were trained with different random seeds (42, 123, 456) and slight variations in audio branch dropout rates (0.35, 0.4, 0.45) to introduce diversity.

## 2.4 Training Process

Training was stable and showed clear convergence. The best single model (seed=123) achieved its highest validation Macro-F1 of 0.8692. The training history (Figure 3) shows that both training and validation metrics improved rapidly in the first 15 epochs before plateauing. Early stopping triggered at epoch 39, indicating efficient training without overfitting. The use of class-balanced sampling and data augmentation contributed to the smooth learning curves.

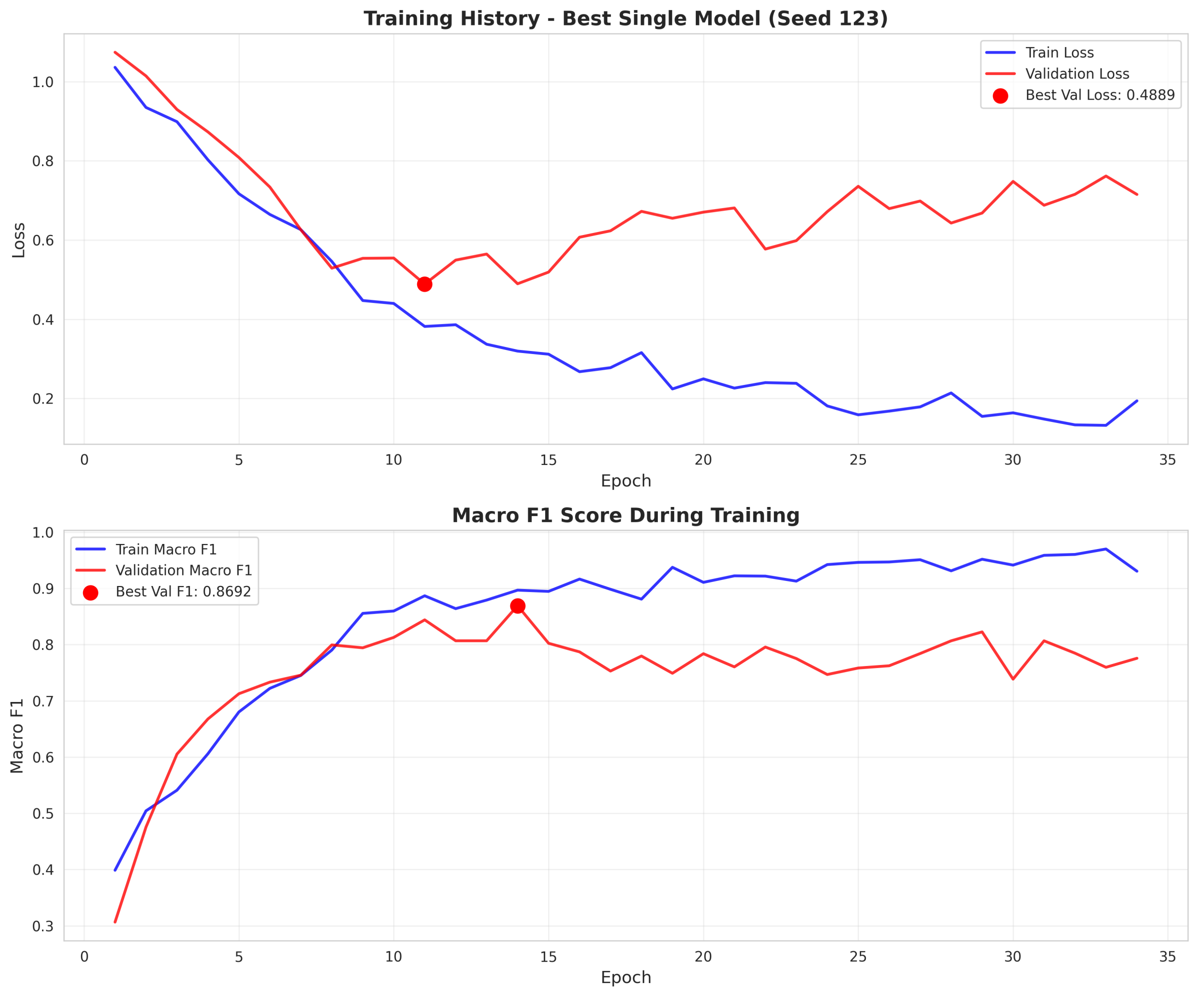


Figure 3: Training history of the best single model. **Top:** Training and validation loss curves, showing steady decrease and convergence. **Bottom:** Training and validation Macro-F1 scores, demonstrating consistent improvement and a final validation F1 of 0.8692. The early stopping point is indicated.

# 3 Model Performance

The final model is a deep ensemble of the three independently trained networks. Their predictions are averaged and calibrated using temperature scaling.

Table 1: Test Set Performance of the Deep Ensemble Model

| Metric | Value |
|---|---|
| Macro-F1 | 0.8384 |
| Accuracy | 0.8545 |
| Weighted F1 | 0.8569 |
| Expected Calibration Error (ECE) | 0.0797 |
| Brier Score | 0.2182 |

The ensemble achieved a test Macro-F1 of **0.8384** and an accuracy of **85.45%**, indicating strong overall performance. The confusion matrix (Figure 4) shows that the recall values for all three tasks are similar (approximately 0.8), demonstrating the effectiveness of the proposed approach in handling imbalanced training data. The low ECE suggests the model's predicted probabilities are well-calibrated, meaning a prediction of 80̈% confidence" is correct approximately 80% of the time.

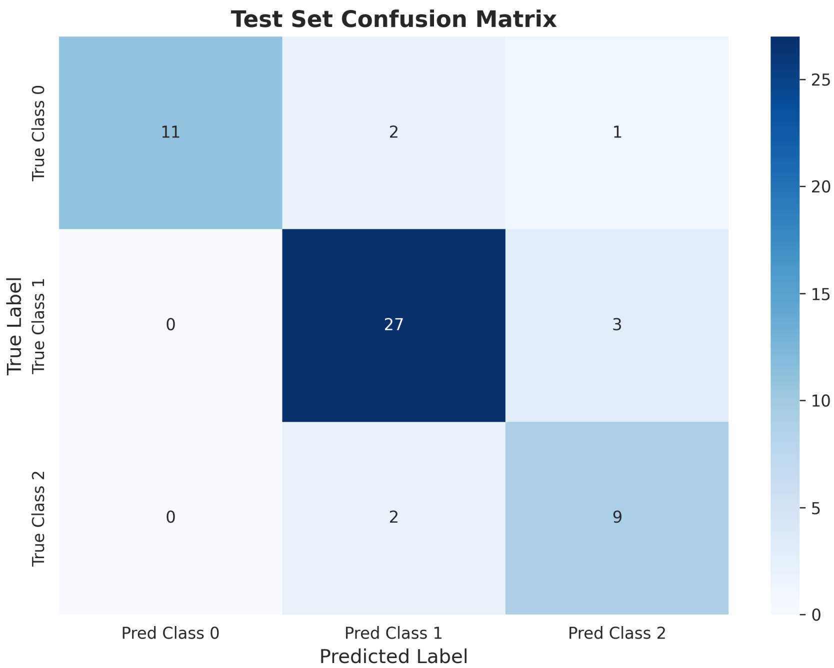


Figure 4: Normalized confusion matrix for the test set predictions. The model performs well on Class 0 and Class 1 (high diagonal values). Class 2 shows more errors, with several instances being misclassified as Class 1, reflecting the difficulty in learning from the smallest class.

# 4 Uncertainty Analysis

The deep ensemble framework provides two types of uncertainty: **Aleatoric** (data noise) and **Epistemic** (model uncertainty). We focus on the total uncertainty, which is the entropy of the ensemble's calibrated predictive distribution.

**Clinical Utility of Uncertainty:** Figure 6 demonstrates a key finding: predictions that are incorrect tend to have higher associated uncertainty. This provides a practical tool for clinical deployment. A clinician can set a threshold on the model's uncertainty score. Predictions with uncertainty below this threshold can be trusted and potentially automated, while high-uncertainty predictions can be flagged for manual review by a specialist. This creates a human-in-the-loop system that improves safety and efficiency.

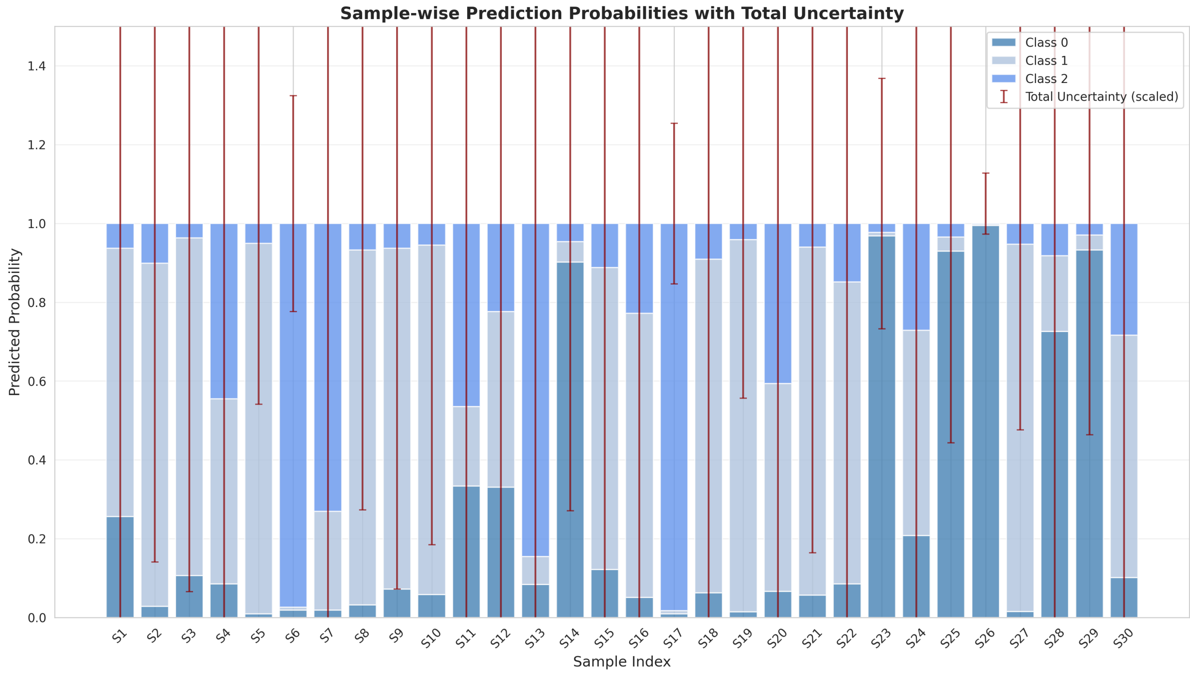


Figure 5: Sample-level uncertainty for 30 test samples. Stacked bars show the predicted probability for each class. The red error bars represent the total uncertainty (entropy). Samples with higher uncertainty (longer error bars) often have more evenly distributed probabilities across classes, indicating lower model confidence.

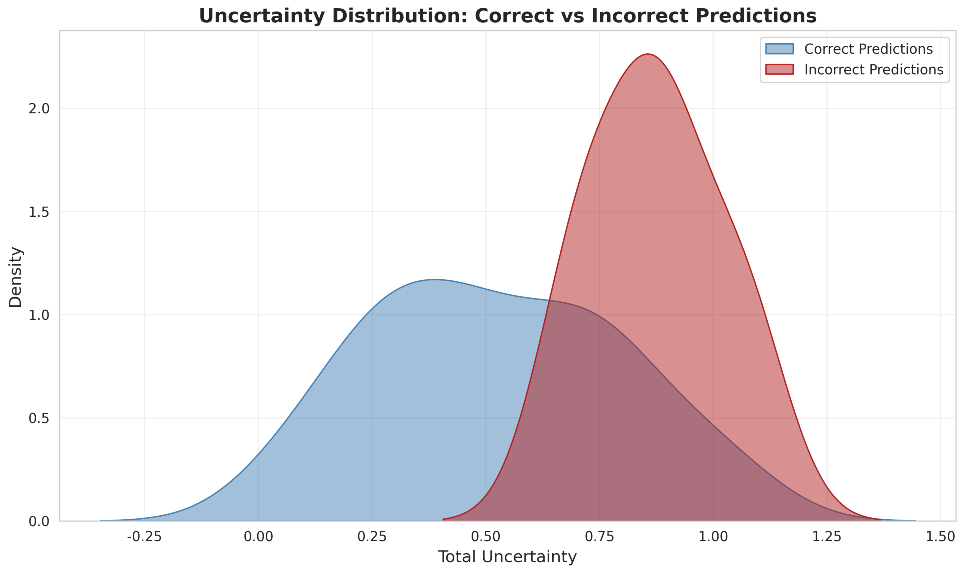


Figure 6: Distribution of total uncertainty for correctly vs. incorrectly predicted samples. Incorrect predictions (red) have a significantly higher average uncertainty than correct predictions (blue). This clear separation means uncertainty can reliably flag predictions that are more likely to be wrong.

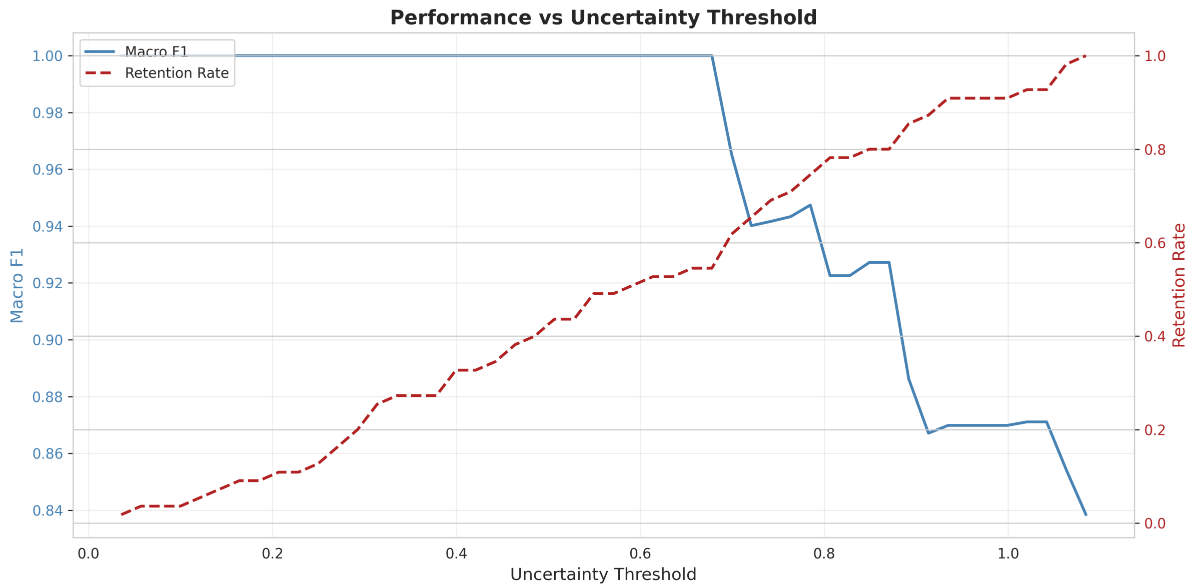


Figure 7: Trade-off between prediction reliability and sample coverage. As the uncertainty threshold increases (allowing more high-uncertainty samples), the Macro-F1 on the retained "confident" samples (blue line) initially stays high but eventually drops. The red line shows the fraction of test samples retained. An optimal operating point can be chosen based on the required confidence level.

Figure 7 quantifies this trade-off. By rejecting samples with high uncertainty, the performance (Macro F1) on the remaining confident subset can be increased above the overall test score, at the cost of not making a prediction for some cases. For example, rejecting the 20% most uncertain samples could yield a system with higher accuracy for its automated decisions.

# 5 Conclusion

This AutoHealth experiment successfully developed a robust multi-modal model for respiratory disease classification using tabular metadata and audio features. The dual-branch deep ensemble achieved a test Macro-F1 of 0.8384, demonstrating effective fusion of complementary data sources.

**Key Findings and Recommendations:**

1. **Model is Effective but Still Can Be Improved**: While the ensemble demonstrates strong and balanced performance, the remaining class confusions indicate potential for further improvement through more discriminative modeling.

2. **Uncertainty as a Safety Filter**: The model's predictive uncertainty is a reliable indicator of potential error. We strongly recommend implementing an uncertainty threshold in any deployment pipeline. Predictions with low uncertainty can be acted upon with confidence, while high-uncertainty cases should be reviewed by a clinician.

3. **Data Quality is Foundational**: The preprocessing steps addressing missing values, embedding inconsistency, and class imbalance were crucial for this performance. Future data collection should aim to reduce these issues at the source.

4. **Actionable Guidance**: For clinical use, we propose a two-tier system: 1) **Automated Triage**: The model provides a diagnosis and an uncertainty score. 2) **Clinician Review**: Cases with high uncertainty or a predicted label of Class 2 are automatically routed for expert review. This balances efficiency with diagnostic safety.

The developed model and its associated uncertainty estimates provide a solid foundation for a decision-support tool that can enhance the efficiency and accuracy of respiratory disease screening.